

Creation Date 20-Aug-2014

Revision Date 22-Mar-2024

Revision Number 2

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                                           |
|---------------------------|-----------------------------------------------------------|
| Product Description:      | <b>Mercury, Triple Distilled 99.99999% (metals basis)</b> |
| Cat No. :                 | <b>98904</b>                                              |
| Synonyms                  | Quicksilver                                               |
| Index No                  | 080-001-00-0                                              |
| CAS No                    | 7439-97-6                                                 |
| EC No                     | 231-106-7                                                 |
| Molecular Formula         | Hg                                                        |
| REACH registration number | -                                                         |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Based on available data, the classification criteria are not met

### Health hazards

Acute Inhalation Toxicity - Vapors

Reproductive Toxicity

Specific target organ toxicity - (repeated exposure)

Category 2 (H330)

Category 1B (H360D)

Category 1 (H372)

### Environmental hazards

Acute aquatic toxicity

Chronic aquatic toxicity

Category 1 (H400)

Category 1 (H410)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### **Hazard Statements**

H330 - Fatal if inhaled

H360D - May damage the unborn child

H372 - Causes damage to organs through prolonged or repeated exposure

H410 - Very toxic to aquatic life with long lasting effects

### **Precautionary Statements**

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P310 - Immediately call a POISON CENTER or doctor/physician

P273 - Avoid release to the environment

### **Additional EU labelling**

Restricted to professional users

## 2.3. Other hazards

Toxicity to Soil Dwelling Organisms

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

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## 3.1. Substances

| Component | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                 |
|-----------|-----------|-------------------|----------|-------------------------------------------------------------------------------------------------------------------|
| Mercury   | 7439-97-6 | EEC No. 231-106-7 | 100      | Acute Tox. 2 (H330)<br>Repr. 1B (H360D)<br>STOT RE 1 (H372)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) |

| Component | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-----------|---------------------------------------|----------|-----------------|
| Mercury   | STOT RE 2 (H373) :: C>=0.1%           | 100      | -               |

### Note

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

|                           |   |
|---------------------------|---|
| REACH registration number | - |
|---------------------------|---|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| Ingestion                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

Neurological disorders. May cause central nervous system depression: May cause adverse kidney effects: May cause adverse liver effects: Symptoms may be delayed: Chronic exposure damages the brain and the central nervous system

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Notes to Physician | Treat symptomatically. |
|--------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

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## 5.1. Extinguishing media

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

## 5.2. Special hazards arising from the substance or mixture

Very toxic. Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes. Keep product and empty container away from heat and sources of ignition. Do not allow run-off from fire-fighting to enter drains or water courses.

### **Hazardous Combustion Products**

Mercury oxide, Toxic fumes.

## 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store in metal containers.

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Technical Rules for Hazardous Substances (TRGS) 510  
Storage Class (LGK) (Germany)

Storage Class/LGK 6.1B

Switzerland - Storage of hazardous substances

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union                   | The United Kingdom               | France                                                                                                               | Belgium                                    | Spain                                          |
|-----------|----------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------|
| Mercury   | TWA: 0.02 mg/m <sup>3</sup> (8h) | TWA: 0.02 mg/m <sup>3</sup> 8 hr | TWA / VME: 0.02 mg/m <sup>3</sup> (8 heures). restrictive limit TWA / VME: 0.1 mg/m <sup>3</sup> (8 heures).<br>Peau | TWA: 0.02 mg/m <sup>3</sup> 8 uren<br>Huid | TWA / VLA-ED: 0.02 mg/m <sup>3</sup> (8 horas) |

| Component | Italy                                                                | Germany                                                                                                                                                       | Portugal                                    | The Netherlands                    | Finland                                       |
|-----------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|------------------------------------|-----------------------------------------------|
| Mercury   | TWA: 0.02 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>Pelle | TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.16 mg/m <sup>3</sup><br>Haut | TWA: 0.02 mg/m <sup>3</sup> 8 horas<br>Pele | TWA: 0.02 mg/m <sup>3</sup> 8 uren | TWA: 0.02 mg/m <sup>3</sup> 8 tunteina<br>Iho |

| Component | Austria                                                                                          | Denmark                                                                                | Switzerland                                                                                                                                                                                                                               | Poland                                  | Norway                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------|
| Mercury   | Haut<br>MAK-KZGW: 0.08 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 0.02 mg/m <sup>3</sup> 8 Stunden | TWA: 0.02 mg/m <sup>3</sup> 8 timer<br>STEL: 0.04 mg/m <sup>3</sup> 15 minutter<br>Hud | Haut/Peau<br>STEL: 0.04 ppm 15 Minuten<br>STEL: 0.4 mg/m <sup>3</sup> 15 Minuten<br>STEL: 0.16 mg/m <sup>3</sup> 15 Minuten<br>TWA: 0.005 ppm 8 Stunden<br>TWA: 0.05 mg/m <sup>3</sup> 8 Stunden<br>TWA: 0.02 mg/m <sup>3</sup> 8 Stunden | TWA: 0.02 mg/m <sup>3</sup> 8 godzinach | TWA: 0.02 mg/m <sup>3</sup> 8 timer<br>STEL: 0.06 mg/m <sup>3</sup> 15 minutter. value calculated |

| Component | Bulgaria                                                   | Croatia                                                                                                                                                                  | Ireland                                                                        | Cyprus                      | Czech Republic                                                                                                                             |
|-----------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Mercury   | TWA: 0.05 mg/m <sup>3</sup><br>TWA: 0.02 mg/m <sup>3</sup> | TWA-GVI: 0.02 mg/m <sup>3</sup> 8 satima. relevant biological monitoring techniques to supplement the indicative occupational exposure limit values should be considered | TWA: 0.02 mg/m <sup>3</sup> 8 hr.<br>Hg<br>STEL: 0.06 mg/m <sup>3</sup> 15 min | TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 0.15 mg/m <sup>3</sup><br>toxic for reproduction |

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|  |  |                                                                                |  |  |  |
|--|--|--------------------------------------------------------------------------------|--|--|--|
|  |  | during the monitoring of mercury exposure and its divalent inorganic compounds |  |  |  |
|--|--|--------------------------------------------------------------------------------|--|--|--|

| Component | Estonia                                 | Gibraltar                                                                                                                                                                                                       | Greece                      | Hungary                                                                          | Iceland                                                                                    |
|-----------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Mercury   | TWA: 0.02 mg/m <sup>3</sup> 8 tundides. | TWA: 0.02 mg/m <sup>3</sup> 8 hr during exposure monitoring for mercury and its divalent inorganic compounds, account should be taken of relevant biological monitoring techniques that complement the IOELV Hg | TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> 8 órában. AK lehetséges borön keresztüli felszívódás | TWA: 0.02 mg/m <sup>3</sup> 8 klukkustundum. Skin notation Ceiling: 0.04 mg/m <sup>3</sup> |

| Component | Latvia                      | Lithuania                        | Luxembourg                            | Malta                       | Romania                           |
|-----------|-----------------------------|----------------------------------|---------------------------------------|-----------------------------|-----------------------------------|
| Mercury   | TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> IPRD | TWA: 0.02 mg/m <sup>3</sup> 8 Stunden | TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> 8 ore |

| Component | Russia                                                           | Slovak Republic                                                  | Slovenia                                                                                                                    | Sweden                                    | Turkey |
|-----------|------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------|
| Mercury   | TWA: 0.005 mg/m <sup>3</sup> 1808<br>MAC: 0.01 mg/m <sup>3</sup> | Potential for cutaneous absorption<br>TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> 8 urah inhalable fraction<br>Koža<br>STEL: 0.16 mg/m <sup>3</sup> 15 minutah inhalable fraction | TLV: 0.02 mg/m <sup>3</sup> 8 timmar. NGV |        |

## Biological limit values

List source(s): **UK** - Biological Monitoring Guidance Values provided by the UK's Health and Safety Executive (HSE) Control of Substances Hazardous to Health Regulations (COSHH) 2002 (as amended) and EH40/2005.

| Component | European Union | United Kingdom                               | France                                                                                                                                           | Spain                                                                                                                 | Germany                                             |
|-----------|----------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Mercury   |                | Mercury: 20 µmol/mol creatinine urine random | Total inorganic Mercury: 0.015 mg/L blood end of shift at end of workweek<br>Total inorganic Mercury: 0.050 mg/g creatinine urine prior to shift | total inorganic mercury: 30 µg/g Creatinine urine pre-shift<br>total inorganic mercury: 10 µg/L blood end of workweek | Mercury: 25 µg/g Creatinine urine (no restriction ) |

| Component | Italy | Finland                                                                                                                                                                                                         | Denmark | Bulgaria                                                         | Romania                                                                                          |
|-----------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Mercury   |       | Mercury: 140 nmol/L urine in the morning after a working day at the end of a working week or exposure period.<br>Mercury, inorganic: 50 nmol/L blood at the end of a working week; time of day does not matter. |         | Mercury: 100 µg/L urine not fixed metal vapor in elemental state | Mercury: 10 µg/L blood end of shift<br>Mercury: 30 µg/g Creatinine urine beginning of next shift |

| Component | Gibraltar | Latvia                                                      | Slovak Republic                                                                                              | Luxembourg | Turkey |
|-----------|-----------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------|--------|
| Mercury   |           | Mercury: 10 µg/L blood<br>Mercury: 30 µg/g Creatinine urine | Mercury: 37.5 µg/L urine not critical<br>Mercury: 15 mg/L blood after all work shifts for long-term exposure |            |        |

## Monitoring methods

MDHS16/2 Mercury and its inorganic divalent compounds in air Laboratory method using Hydrar diffusive badges or pumped sorbent tubes, acid dissolution and analysis by cold vapour atomic absorption spectrometry or cold vapour atomic fluorescence spectrometry

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## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                    | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Mercury<br>7439-97-6 ( 100 ) |                                  |                                     |                                    | DNEL = 0.02mg/m <sup>3</sup>          |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                    | Fresh water       | Fresh water sediment        | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)     |
|------------------------------|-------------------|-----------------------------|--------------------|------------------------------------|------------------------|
| Mercury<br>7439-97-6 ( 100 ) | PNEC = 0.0574µg/L | PNEC = 9.3mg/kg sediment dw |                    | PNEC = 2.25µg/L                    | PNEC = 22µg/kg soil dw |

| Component                    | Marine water      | Marine water sediment       | Marine water Intermittent | Food chain | Air |
|------------------------------|-------------------|-----------------------------|---------------------------|------------|-----|
| Mercury<br>7439-97-6 ( 100 ) | PNEC = 0.0672µg/L | PNEC = 9.3mg/kg sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | > 480 minutes     | 0.54mm          | Level 6     | (minimum requirement) |
| Natural rubber | > 480 minutes     | 0.48mm          | EN 374      |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particulates filter conforming to EN 143 or Inorganic gases and vapours filter Type B Grey conforming to EN14387

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure

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limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Liquid                   |                                   |
| Appearance                              | Silver                   |                                   |
| Odor                                    | Odorless                 |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | -38.9 °C / -38 °F        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | 356.5 °C / 673.7 °F      |                                   |
| Flammability (liquid)                   | No data available        |                                   |
| Flammability (solid,gas)                | Not applicable           | Liquid                            |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | Not applicable           |                                   |
| Viscosity                               | 1.554 cP at 20 °C        |                                   |
| Water Solubility                        | Insoluble                |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Vapor Pressure                          | 0.01 hPa @ 20 °C         |                                   |
| Density / Specific Gravity              | 13.540                   |                                   |
| Bulk Density                            | Not applicable           | Liquid                            |
| Vapor Density                           | 7.0                      | (Air = 1.0)                       |
| Particle characteristics                | Not applicable (liquid)  |                                   |

### 9.2. Other information

|                   |        |
|-------------------|--------|
| Molecular Formula | Hg     |
| Molecular Weight  | 200.59 |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

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## 10.5. Incompatible materials

Strong oxidizing agents. Ammonia. Metals. Halogens.

## 10.6. Hazardous decomposition products

Mercury oxide. Toxic fumes.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

|            |                   |
|------------|-------------------|
| Oral       | No data available |
| Dermal     | No data available |
| Inhalation | Category 2        |

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation                         |
|-----------|-----------|-------------|-----------------------------------------|
| Mercury   | -         | -           | LC50 < 27 mg/m <sup>3</sup> ( Rat ) 2 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

##### (g) reproductive toxicity; Developmental Effects

Category 1B  
May cause harm to the unborn child.

(h) STOT-single exposure; No data available

##### (i) STOT-repeated exposure;

|               |                                              |
|---------------|----------------------------------------------|
| Target Organs | Kidney, Liver, Central nervous system (CNS). |
|---------------|----------------------------------------------|

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** May cause central nervous system depression. May cause adverse kidney effects. May cause adverse liver effects. Symptoms may be delayed. Chronic exposure damages the brain and the central nervous system.

### 11.2. Information on other hazards

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**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### **Ecotoxicity effects**

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                                                    | Water Flea | Freshwater Algae |
|-----------|------------------------------------------------------------------------------------|------------|------------------|
| Mercury   | 0.9 mg/L LC50 96h<br>0.18 mg/L LC50 96h<br>0.16 mg/L LC50 96h<br>0.5 mg/L LC50 96h |            |                  |

| Component | Microtox | M-Factor |
|-----------|----------|----------|
| Mercury   |          | 100      |

### 12.2. Persistence and degradability

**Persistence**  
**Degradability**  
**Degradation in sewage treatment plant**

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
Insoluble in water, May persist.  
Not relevant for inorganic substances.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

Product has a high potential to bioconcentrate

### 12.4. Mobility in soil

Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

#### **Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

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|                                       |                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>European Waste Catalogue (EWC)</b> | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>              | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.                                                                                                       |
| <b>Switzerland - Waste Ordinance</b>  | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                         |         |
|-----------------------------------------|---------|
| <b>14.1. UN number</b>                  | UN2809  |
| <b>14.2. UN proper shipping name</b>    | Mercury |
| <b>14.3. Transport hazard class(es)</b> | 8       |
| <b>Subsidiary Hazard Class</b>          | 6.1     |
| <b>14.4. Packing group</b>              | III     |

### ADR

|                                         |         |
|-----------------------------------------|---------|
| <b>14.1. UN number</b>                  | UN2809  |
| <b>14.2. UN proper shipping name</b>    | Mercury |
| <b>14.3. Transport hazard class(es)</b> | 8       |
| <b>Subsidiary Hazard Class</b>          | 6.1     |
| <b>14.4. Packing group</b>              | III     |

### IATA

|                                         |         |
|-----------------------------------------|---------|
| <b>14.1. UN number</b>                  | UN2809  |
| <b>14.2. UN proper shipping name</b>    | Mercury |
| <b>14.3. Transport hazard class(es)</b> | 8       |
| <b>Subsidiary Hazard Class</b>          | 6.1     |
| <b>14.4. Packing group</b>              | III     |

|                                    |                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>14.5. Environmental hazards</b> | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
|------------------------------------|----------------------------------------------------------------------------------------------------------|

|                                           |                                  |
|-------------------------------------------|----------------------------------|
| <b>14.6. Special precautions for user</b> | No special precautions required. |
|-------------------------------------------|----------------------------------|

|                                                                      |                                |
|----------------------------------------------------------------------|--------------------------------|
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods |
|----------------------------------------------------------------------|--------------------------------|

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Mercury   | 7439-97-6 | 231-106-7 | -      | -   | X     | X    | KE-23117 | X    | -    |

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| Component | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Mercury   | 7439-97-6 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                     | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Mercury   | 7439-97-6 | -                                                                   | Use restricted. See item 18[a].<br>(see link for restriction details)<br>Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Mercury   | 7439-97-6 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

| Component                    | ANNEX I - PART 1<br>List of chemicals subject to export notification procedure (referred to in Article 8)                                                                                                                                                                                                                                                                                                                                      | ANNEX I - PART 2<br>List of chemicals qualifying for PIC notification (referred to in Article 11) | ANNEX I - PART 3<br>List of chemicals subject to the PIC procedure (referred to in Articles 13 and 14) |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Mercury<br>7439-97-6 ( 100 ) | p(1) — pesticide in the group of plant protection products<br>b — ban (for the category or categories concerned)<br><br>p(2) — other pesticide including biocides<br>b — ban (for the category or categories concerned)<br><br>i(1) — industrial chemical for professional use<br>i(2) — industrial chemical for public<br>sr — severe restriction<br><br>Ref — Please refer to PIC circular at <a href="http://www.pic.int/">www.pic.int/</a> | i — industrial chemical<br>sr — severe restriction                                                | p — pesticides                                                                                         |

<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32012R0649&qid=1604065742303>.

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## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                |
|-----------|---------------------------------------|--------------------------------------------------------|
| Mercury   | WGK3                                  | Class I : 0.01 mg/m <sup>3</sup> (Massenkonzentration) |

| Component | France - INRS (Tables of occupational diseases)     |
|-----------|-----------------------------------------------------|
| Mercury   | Tableaux des maladies professionnelles (TMP) - RG 2 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                    | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Mercury<br>7439-97-6 ( 100 ) | Prohibited and Restricted Substances                                                                           |                                                                                 | Annex I - pesticide<br>Annex I - industrial chemical<br>Annex II - pesticide                |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H330 - Fatal if inhaled

H360D - May damage the unborn child

H372 - Causes damage to organs through prolonged or repeated exposure

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

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**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

## Prepared By

Health, Safety and Environmental Department

## Creation Date

20-Aug-2014

## Revision Date

22-Mar-2024

## Revision Summary

New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

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**End of Safety Data Sheet**